

EDS Lab Assignments

Practical 3

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3.1.1 Numpy Array Operations

Problem Statement:

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`
- The shape of the array using `shape`
- The total number of elements in the array using `size`

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers `r` and `c`, representing the number of rows and the number of columns.
- The next `r` lines contain `c` space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Code:

```
import numpy as np
rows, cols = map(int, input().split())
elements = []

for _ in range (rows):
    elements.extend(map(int, input().split()))

array = np.array(elements).reshape(rows, cols)
print(array)
print(array.ndim)
print(array.shape)
print(array.size)
```

Output:

Average time
0.010 s
9.75 ms


0.017 s
17.00 ms

✓ 2 out of 2 shown test case(s) passed

✓ 10 out of 10 hidden test case(s) passed

✓ Test case 1 17 ms

Debug

Expected output

Actual output

3 4	3 4
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
9 10 11 12	9 10 11 12
[[·1··2··3··4]	[[·1··2··3··4]
·[·5··6··7··8]	·[·5··6··7··8]
·[·9·10·11·12]]	·[·9·10·11·12]]
2	2
(3, -4)	(3, -4)
12	12

✓ Test case 2 6 ms

Debug

Expected output

Actual output

0 0	0 0
[]	[]
2	2
(0, -0)	(0, -0)
0	0

3.2.1 Numpy: Matrix Operations

Problem Statement:

The given code takes two 3×3 matrices, `matrix_a`, and `matrix_b`, as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. **Addition** (`matrix_a + matrix_b`)
2. **Subtraction** (`matrix_a - matrix_b`)
3. **Element-wise Multiplication** (`matrix_a * matrix_b`)
4. **Matrix Multiplication** (`matrix_a · matrix_b`)
5. **Transpose of Matrix A**

Input Format:

- The user will input 3 rows for `matrix_a`, each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for `matrix_b`, each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

Code:

```
import numpy as np

# Input matrices
print("Enter Matrix A:")
matrix_a = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Matrix B:")
matrix_b = np.array([list(map(int, input().split())) for i in range(3)])

# Addition
addition_result = matrix_a + matrix_b
print("Addition (A + B):")
print(addition_result)

# Subtraction
subtraction_result = matrix_a - matrix_b
print("Subtraction (A - B):")
print(subtraction_result)

# Multiplication (element-wise)
elementwise_multiplication_result = matrix_a * matrix_b
print("Element-wise Multiplication (A * B):")
print(elementwise_multiplication_result)

# Matrix multiplication (dot product)
matrix_multiplication_result = np.dot(matrix_a, matrix_b)
print("A dot B:")
print(matrix_multiplication_result)

# Transpose
transpose_a = matrix_a.T
print("Transpose of A:")
print(transpose_a)
```

Output:

Average time
0.019 s
18.75 ms


0.024 s
24.00 ms

✓ 2 out of 2 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 24 ms  Debug  

Expected output

Actual output

Enter Matrix A:	Enter Matrix A:
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Matrix B:	Enter Matrix B:
1 1 1	1 1 1
1 1 1	1 1 1
1 1 1	1 1 1
Addition (A+B):	Addition (A+B):
[[2 3 4]	[[2 3 4]
5 6 7]	5 6 7]
8 9 10]]	8 9 10]]
Subtraction (A-B):	Subtraction (A-B):
[[0 1 2]	[[0 1 2]
3 4 5]	3 4 5]
6 7 8]]	6 7 8]]
Element-wise Multiplication (A*B):	Element-wise Multiplication (A*B):
[[1 2 3]	[[1 2 3]
4 5 6]	4 5 6]

Average time

0.019 s

18.75 ms



Maximum time

0.024 s

24.00 ms



✓ 2 out of 2 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

· [7 · 8 · 9]]

· [7 · 8 · 9]]

A · dot · B:

A · dot · B:

[[· 6 · · 6 · · 6]

[[· 6 · · 6 · · 6]

· [15 · 15 · 15]

· [15 · 15 · 15]

· [24 · 24 · 24]]

· [24 · 24 · 24]]

Transpose · of · A:

Transpose · of · A:

[[1 · 4 · 7]

[[1 · 4 · 7]

· [2 · 5 · 8]

· [2 · 5 · 8]

· [3 · 6 · 9]]

· [3 · 6 · 9]]

✓ Test case 2 20 ms

Debug



Expected output

Actual output

Enter · Matrix · A:

Enter · Matrix · A:

2 4 8

2 4 8

9 6 5

9 6 5

7 8 9

7 8 9

Enter · Matrix · B:

Enter · Matrix · B:

10 12 13

10 12 13

14 15 16

14 15 16

17 18 19

17 18 19

Addition · (A · + · B):

Addition · (A · + · B):

Addition · (A + B):	Addition · (A + B):
[[12 · 16 · 21]	[[12 · 16 · 21]
· [23 · 21 · 21]	· [23 · 21 · 21]
· [24 · 26 · 28]]	· [24 · 26 · 28]]
Subtraction · (A - B):	Subtraction · (A - B):
[[· -8 · -8 · -5]	[[· -8 · -8 · -5]
· [· -5 · -9 · -11]	· [· -5 · -9 · -11]
· [-10 · -10 · -10]]	· [-10 · -10 · -10]]
Element-wise Multiplication · (A * B):	Element-wise Multiplication · (A * B):
[[· 20 · 48 · 104]	[[· 20 · 48 · 104]
· [126 · 90 · 80]	· [126 · 90 · 80]
· [119 · 144 · 171]]	· [119 · 144 · 171]]
A · dot · B:	A · dot · B:
[[212 · 228 · 242]	[[212 · 228 · 242]
· [259 · 288 · 308]	· [259 · 288 · 308]
· [335 · 366 · 390]]	· [335 · 366 · 390]]
Transpose of A:	Transpose of A:
[[2 · 9 · 7]	[[2 · 9 · 7]
· [4 · 6 · 8]	· [4 · 6 · 8]
· [8 · 5 · 9]]	· [8 · 5 · 9]]

3.2.2 Numpy: Horizontal and Vertical Stacking of Arrays

Problem Statement:

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking:** Stack the two matrices horizontally (side by side).
- **Vertical Stacking:** Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Code:

```
import numpy as np

# Input matrices
print("Enter Array1:")
arr1 = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Array2:")
arr2 = np.array([list(map(int, input().split())) for i in range(3)])

# Perform horizontal stacking (hstack)
a = np.hstack((arr1, arr2))
print("Horizontal Stack:")
print(a)

# Perform vertical stacking (vstack)
b = np.vstack((arr1, arr2))
print("Vertical Stack:")
print(b)
```

Output:

Average time
0.018 s
18.50 ms

Maximum time
0.024 s
24.00 ms

✓ 2 out of 2 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 24 ms Debug

Expected output

Actual output

Enter Array1:
1 2 3
4 5 6
7 8 9
Enter Array2:
4 5 6
7 8 9
10 11 12
Horizontal Stack:
[[1 2 3 4 5 6]
[4 5 6 7 8 9]
[7 8 9 10 11 12]]
Vertical Stack:
[[1 2 3]
[4 5 6]
[7 8 9]
[4 5 6]
[7 8 9]
[10 11 12]]

Enter Array1:
1 2 3
4 5 6
7 8 9
Enter Array2:
4 5 6
7 8 9
10 11 12
Horizontal Stack:
[[1 2 3 4 5 6]
[4 5 6 7 8 9]
[7 8 9 10 11 12]]
Vertical Stack:
[[1 2 3]
[4 5 6]
[7 8 9]
[4 5 6]
[7 8 9]
[10 11 12]]

3.2.3 Numpy: Custom Sequence Generation

Problem Statement:

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

Code:

```
import numpy as np
# Take user input for the start, stop, and step of the sequence
start = int(input())
stop = int(input())
step = int(input())
# Generate the sequence using np.arange()
a = np.arange(start, stop, step)
# Print the generated sequence
print(a)
```

Output:

Average time
0.009 s
9.50 ms

Maximum time
0.015 s
15.00 ms

✓ 2 out of 2 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 15 ms Debug

Expected output	Actual output
3	3
10	10
2	2
[3·5·7·9]	[3·5·7·9]

✓ Test case 2 8 ms Debug

Expected output	Actual output
10	10
20	20
30	30
[10]	[10]

3.2.4 Numpy: Arithmetic and Statistical Operations, Mathematical Operations and Bitwise Operators

Problem Statement:

You are given two arrays A and B. Your task is to complete the function `array_operations`, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

- Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

- Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_i OR B_i).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Code:

```
import numpy as np

def array_operations(A, B):
    # Convert A and B to NumPy arrays
    A = np.array(A)
    B = np.array(B)

    # Arithmetic Operations
    sum_result = A + B
    diff_result = A - B
    prod_result = A * B

    # Statistical Operations
    mean_A = np.mean(A)
    median_A = np.median(A)
```

```

std_dev_A = np.std(A)

# Bitwise Operations
and_result = A & B
or_result = A | B
xor_result = A ^ B

# Output results with one space between each element
print("Element-wise Sum:", ' '.join(map(str, sum_result)))
print("Element-wise Difference:", ' '.join(map(str, diff_result)))
print("Element-wise Product:", ' '.join(map(str, prod_result)))

print(f"Mean of A: {mean_A}")
print(f"Median of A: {median_A}")
print(f"Standard Deviation of A: {std_dev_A}")
    print("Bitwise AND:", ' '.join(map(str, and_result)))
print("Bitwise OR:", ' '.join(map(str, or_result)))
print("Bitwise XOR:", ' '.join(map(str, xor_result)))

A = list(map(int, input().split())) # Elements of array A
B = list(map(int, input().split())) # Elements of array B
array_operations(A, B)

```

Output:

Average time

0.011 s

10.67 ms

Maximum time

0.014 s

14.00 ms

✓ 1 out of 1 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1

14 ms

Debug

Expected output	Actual output
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
Element-wise Sum: ·6·8·10·12	Element-wise Sum: ·6·8·10·12
Element-wise Difference: ·-4·-4·-4·-4	Element-wise Difference: ·-4·-4·-4·-4
Element-wise Product: ·5·12·21·32	Element-wise Product: ·5·12·21·32
Mean of A: ·2.5	Mean of A: ·2.5
Median of A: ·2.5	Median of A: ·2.5
Standard Deviation of A: ·1.118033988749895	Standard Deviation of A: ·1.118033988749895
Bitwise AND: ·1·2·3·0	Bitwise AND: ·1·2·3·0
Bitwise OR: ·5·6·7·12	Bitwise OR: ·5·6·7·12
Bitwise XOR: ·4·4·4·12	Bitwise XOR: ·4·4·4·12

3.2.5 Numpy: Copying and Viewing Arrays

Problem Statement:

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the original_array and assigning it to view_array.
- Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:
Original array after modifying view: <original_array>
View array: <view_array>
- After modifying the copy:
Original array after modifying copy: <original_array>
Copy array: <copy_array>

Code:

```
import numpy as np
inputlist = list(map(int,input().split(" ")))

# Original array
original_array = np.array(inputlist)

# Create a view
view_array = original_array.view()

# Create a copy
copy_array = original_array.copy()

# Modify the view
view_array[0] = 99
print("Original array after modifying view:", original_array)
print("View array:", view_array)

# Modify the copy
copy_array[1] = 88
print("Original array after modifying copy:", original_array)
print("Copy array:", copy_array)
```

Output:

Average time
0.005 s
5.50 ms


0.008 s
8.00 ms

✓ 2 out of 2 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 8 ms Debug

Expected output

Actual output

10 20 30 40 50 60 70 80

10 20 30 40 50 60 70 80

Original array after modifying view: [99 20 30 40 50 60 70 80]

Original array after modifying view: [99 20 30 40 50 60 70 80]

View array: [99 20 30 40 50 60 70 80]

View array: [99 20 30 40 50 60 70 80]

Original array after modifying copy: [99 20 30 40 50 60 70 80]

Original array after modifying copy: [99 20 30 40 50 60 70 80]

Copy array: [10 88 30 40 50 60 70 80]

Copy array: [10 88 30 40 50 60 70 80]

✓ Test case 2 5 ms Debug

Expected output

Actual output

99 2 3 4 5

99 2 3 4 5

Original array after modifying view: [99 2 3 4 5]

Original array after modifying view: [99 2 3 4 5]

View array: [99 2 3 4 5]

View array: [99 2 3 4 5]

Original array after modifying copy: [99 2 3 4 5]

Original array after modifying copy: [99 2 3 4 5]

Copy array: [99 88 3 4 5]

Copy array: [99 88 3 4 5]

3.2.6 Searching, Sorting, Counting, Broadcasting

Problem Statement:

The given code in the editor takes a single array, array1, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- search_value: The value to search for in the array.
- count_value: The value to count its occurrences in the array.
- broadcast_value: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

- 1. Searching:** Find the indices where search_value appears in array1 and print these indices.
- 2. Counting:** Count how many times count_value appears in array1 and print the count.
- 3. Broadcasting:** Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.
- 4. Sorting:** Sort array1 in ascending order and print the sorted array.

Input Format:

1. A single line containing space-separated integers representing array1.
2. An integer search_value represents the value to search for in the array.
3. An integer count_value represents the value to count in the array.
4. An integer broadcast_value represents the value to add to each element of the array.

Output Format:

1. The indices where search_value occurs in array1.
2. The count of occurrences of count_value in array1.
3. The array after adding the broadcast_value to each element.
4. The sorted array.

Code:

```
import numpy as np

# Input array from the user
array1 = np.array(list(map(int, input().split()))))

# Searching
search_value = int(input("Value to search: "))
count_value = int(input("Value to count: "))
broadcast_value = int(input("Value to add: "))

# Find indices where value matches in array1
a = np.where(array1 == search_value)[0]
print(a)

# Count occurrences in array1
b = np.count_nonzero(array1 == count_value)
print(b)

# Broadcasting addition
c = array1 + broadcast_value
print(c)

# Sort the first array
d = np.sort(array1)
print(d)
```


Output:

Average time
0.015 s
14.50 ms

Maximum time
0.019 s
19.00 ms

✓ 2 out of 2 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 19 ms Debug

Expected output

Actual output

1 1 1 2 2 2

1 1 1 2 2 2

Value to search: 1

Value to search: 1

Value to count: 2

Value to count: 2

Value to add: 2

Value to add: 2

[0 1 2]

[0 1 2]

3

3

[3 3 3 4 4 4]

[3 3 3 4 4 4]

[1 1 1 2 2 2]

[1 1 1 2 2 2]

✓ Test case 2 14 ms Debug

Expected output

Actual output

1 2 3 4 5

1 2 3 4 5

Value to search: 6

Value to search: 6

Value to count: 6

Value to count: 6

Value to add: 6

Value to add: 6

[]

[]

0

0

[-7 -8 -9 10 11]

[-7 -8 -9 10 11]

[1 2 3 4 5]

[1 2 3 4 5]

3.2.7 Student Data Analysis and Operations

Problem Statement:

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details:** Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students:** Determine the total number of students in the dataset.
- **Print all student roll numbers:** Extract and print the roll numbers of all students.
- **Print Subject 1 marks:** Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2:** Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3:** Identify the highest marks in Subject 3.
- **Print all subject marks:** Display the marks of all students for each subject.
- **Find total marks of students:** Compute the total marks for each student across all subjects.
- **Find the average marks of each student:** Compute the average marks for each student.
- **Find average marks of each subject:** Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2:** Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3:** Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of the student with maximum marks in Subject 3:** Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2:** Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2:** Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who got less than 40 marks in Subject 1:** Count the number of students who scored less than 40 marks in Subject 1.
- **Find the count of students who got more than 90 marks in Subject 2:** Count the number of students who scored more than 90 marks in Subject 2.
- **Find the count of students who scored ≥ 90 in each subject:** Count the number of students who scored 90 or more marks in each subject.
- **Find the count of subjects in which each student scored ≥ 90 :** Determine how many subjects each student scored 90 or more marks in.
- **Print Subject 1 marks in ascending order:** Sort and print the marks of students in Subject 1 in ascending order.

- **Print students who scored between 50 and 90 in Subject 1:** Display students who scored marks between 50 and 90 in Subject 1.
- **Find index positions of students who scored 79 in Subject 1:** Identify the index positions of students who scored exactly 79 marks in Subject 1.

Code:

```
import numpy as np

a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)

roll = a[:, 0].astype(float)
s1 = a[:, 1]
s2 = a[:, 2]
s3 = a[:, 3]

# 1. Print all student details
print("All student Details:\n", a)

# 2. print total students
print("Total Students:", len(a))

# 3. Print all student Roll numbers
print("All Student Roll Nos", roll)

# 4. Print subject 1 marks
print("Subject 1 Marks", s1)

# 5. print minimum marks of Subject 2
print("Min marks in Subject 2", np.min(s2))

# 6. print maximum marks of Subject 3
print("Max marks in Subject 3", np.max(s3))

# 7. Print All subject marks
print("All subject marks:", a[:, 1:])

# 8. print Total marks of students
total = s1 + s2 + s3
print("Total Marks", total)

# 9. print average marks of each student
print(np.round(total/3, 1))

# 10. print average marks of each subject
print("Average Marks of each subject", np.mean(a[:, 1:], axis = 0))

# 11. print average marks of S1 and S2
```

```

print("Average Marks of S1 and S2", np.mean([s1, s2], axis = 1))

# 12. print average marks of S1 and S3
print("Average Marks of S1 and S3", np.mean([s1, s3], axis = 1))

# 13. print Roll number who got maximum marks in Subject 3
print("Roll no who got maximum marks in Subject 3", float(roll[np.argmax(s3)]))

# 14. print Roll number who got minimum marks in Subject 2
print("Roll no who got minimum marks in Subject 2", float(roll[np.argmin(s2)]))

# 15. print Roll number who got 24 marks in Subject 2
print("Roll no who got 24 marks in Subject 2", roll[s2 == 24].astype(float).reshape(1, -1))

# 16. print count of students who got marks in Subject 1 < 40
print("Count of students who got marks in Subject 1 < 40", np.sum(s1 < 40))

# 17. print count of students who got marks in Subject 2 > 90
print("Count of students who got marks in Subject 2 > 90:", np.sum(s2 > 90))

# 18. print count of students in each subject who got marks >= 90
print("Count of students in each subject who got marks >= 90:", np.sum(a[:, 1:] >= 90, axis = 0))

# 19. print count of subjects in which each student got marks >= 90
print("Roll no:", roll)
print("Count of subjects in which student got marks >= 90:", np.sum(a[:, 1:] >= 90, axis = 1))

# 20. Print S1 marks in ascending order
print(np.sort(s1))

# 21. Print S1 marks >= 50 and <= 90
print(a[(s1 >= 50) & (s1 <= 90)].astype(float))
print(a.astype(float))

# 22. Print the index position of marks 79
print(np.where(s1 == 79))

```

Output:

Average time

0.016 s

16.00 ms



Maximum time

0.016 s

16.00 ms

**✓ 1 out of 1 shown test case(s) passed****✓ Test case 1** **16 ms** **Debug**

Expected output

Actual output

All student Details:

All student Details:

· [[301.·67.·77.·88.]

· [[301.·67.·77.·88.]

· [302.·78.·88.·77.]

· [302.·78.·88.·77.]

· [303.·45.·56.·89.]

· [303.·45.·56.·89.]

· [304.·88.·98.·45.]

· [304.·88.·98.·45.]

· [305.·78.·88.·99.]

· [305.·78.·88.·99.]

· [306.·68.·78.·36.]

· [306.·68.·78.·36.]

· [307.·79.·12.·45.]

· [307.·79.·12.·45.]

· [308.·46.·45.·77.]

· [308.·46.·45.·77.]

· [309.·89.·32.·88.]

· [309.·89.·32.·88.]

· [310.·79.·24.·33.]]

· [310.·79.·24.·33.]]

Total Students: ·10

Total Students: ·10

All Student Roll Nos · [301.·302.·303.·304.·305.·306.·307.·308.·309.·310.]

All Student Roll Nos · [301.·302.·303.·304.·305.·306.·307.·308.·309.·310.]

Subject 1 Marks · [67.·78.·45.·88.·78.·68.·79.·46.·89.·79.]

Subject 1 Marks · [67.·78.·45.·88.·78.·68.·79.·46.·89.·79.]

Min marks in Subject 2 ·12.0

Min marks in Subject 2 ·12.0

Max marks in Subject 3 ·99.0

Max marks in Subject 3 ·99.0

All subject marks: · [[67.·77.·88.]

All subject marks: · [[67.·77.·88.]

· [78.·88.·77.]

· [78.·88.·77.]

Average time

0.016 s

16.00 ms



Maximum time

0.016 s

16.00 ms

 **1 out of 1 shown test case(s) passed**

· [45. · 56. · 89.]

· [45. · 56. · 89.]

· [88. · 98. · 45.]

· [88. · 98. · 45.]

· [78. · 88. · 99.]

· [78. · 88. · 99.]

· [68. · 78. · 36.]

· [68. · 78. · 36.]

· [79. · 12. · 45.]

· [79. · 12. · 45.]

· [46. · 45. · 77.]

· [46. · 45. · 77.]

· [89. · 32. · 88.]

· [89. · 32. · 88.]

· [79. · 24. · 33.]

· [79. · 24. · 33.]

Total Marks · [232. · 243. · 190. · 231. · 265. · 182. · 136. · 168. · 209. · 136.]

Total Marks · [232. · 243. · 190. · 231. · 265. · 182. · 136. · 168. · 209. · 136.]

[77.3 · 81. · · 63.3 · 77. · · 88.3 · 60.7 · 45.3 · 56. · · 69.7 · 45.3]

[77.3 · 81. · · 63.3 · 77. · · 88.3 · 60.7 · 45.3 · 56. · · 69.7 · 45.3]

Average Marks of each subject · [71.7 · 59.8 · 67.7]

Average Marks of each subject · [71.7 · 59.8 · 67.7]

Average Marks of S1 and S2 · [71.7 · 59.8]

Average Marks of S1 and S2 · [71.7 · 59.8]

Average Marks of S1 and S3 · [71.7 · 67.7]

Average Marks of S1 and S3 · [71.7 · 67.7]

Roll no who got maximum marks in Subject 3 · 305.0

Roll no who got maximum marks in Subject 3 · 305.0

Roll no who got minimum marks in Subject 2 · 307.0

Roll no who got minimum marks in Subject 2 · 307.0

Roll no who got 24 marks in Subject 2 · [[310.]]

Roll no who got 24 marks in Subject 2 · [[310.]]

Count of students who got marks in Subject 1 · < · 40 · 0

Count of students who got marks in Subject 1 · < · 40 · 0

Count of students who got marks in Subject 2 · > · 90 · : · 1

Count of students who got marks in Subject 2 · > · 90 · : · 1

Average time

0.016 s

16.00 ms



Maximum time

0.016 s

16.00 ms

**✓ 1 out of 1 shown test case(s) passed**

Count of students in each subject who got marks >= 90: [0 1 1]

Roll no: [301, 302, 303, 304, 305, 306, 307, 308, 309, 310.]

Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]

[45, 46, 67, 68, 78, 78, 79, 79, 88, 89.]

[[301, 67, 77, 88.]

[302, 78, 88, 77.]

[304, 88, 98, 45.]

[305, 78, 88, 99.]

[306, 68, 78, 36.]

[307, 79, 12, 45.]

[309, 89, 32, 88.]

[310, 79, 24, 33.]]

[[301, 67, 77, 88.]

[302, 78, 88, 77.]

[303, 45, 56, 89.]

[304, 88, 98, 45.]

[305, 78, 88, 99.]

[306, 68, 78, 36.]

[307, 79, 12, 45.]

[308, 46, 45, 77.]

Count of students in each subject who got marks >= 90: [0 1 1]

Roll no: [301, 302, 303, 304, 305, 306, 307, 308, 309, 310.]

Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]

[45, 46, 67, 68, 78, 78, 79, 79, 88, 89.]

[[301, 67, 77, 88.]

[302, 78, 88, 77.]

[304, 88, 98, 45.]

[305, 78, 88, 99.]

[306, 68, 78, 36.]

[307, 79, 12, 45.]

[309, 89, 32, 88.]

[310, 79, 24, 33.]]

[[301, 67, 77, 88.]

[302, 78, 88, 77.]

[303, 45, 56, 89.]

[304, 88, 98, 45.]

[305, 78, 88, 99.]

[306, 68, 78, 36.]

[307, 79, 12, 45.]

[308, 46, 45, 77.]